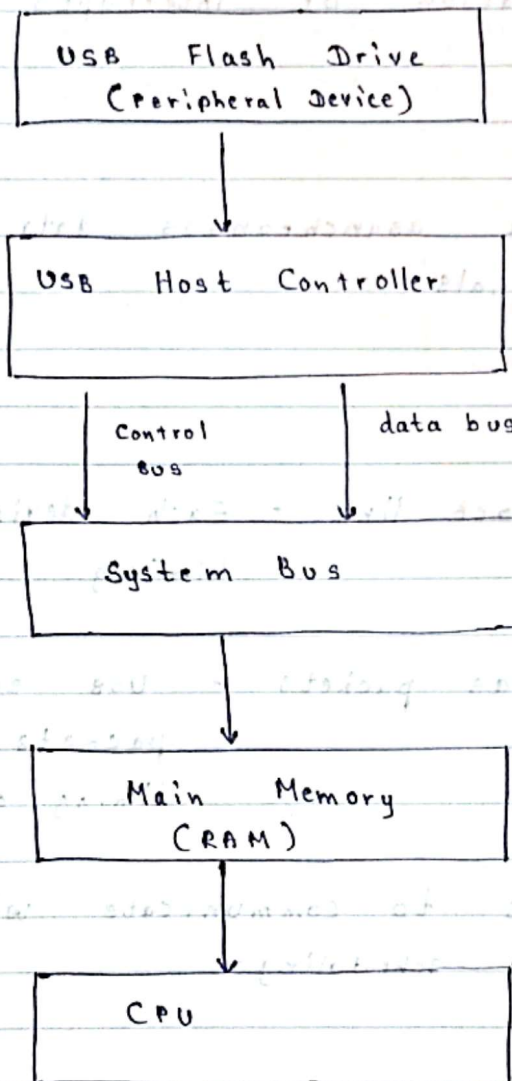


Inclass 04

- 1) The USB Controller is connected to the system bus
 The CPU communicates with the USB host Controller using

Memory-mapped I/O - using standard memory access addresses

I/O - Mapped ports - using In and out instructions.

- The CPU sends commands (read / write requests) to control register in the Controller & receives status information or interrupts from the Controller.

- 2) USB uses a asynchronous data transfer and handshake signals.

Asynchronous

No shared clock line - Each device has its own timing

Data sent as packets - USB sends discrete packets that include timing and control information

- Allow devices to communicate without shared clock.

Hand shake

Reliability check: - After each data packet the receiver sends a handshake packet (ACK, NAK or STALL) to confirm data was received correct

Error detection and retransmission

- Ensure correct delivery and flow control.

use this

3) Initiation type	Initiated by	Used in USB	Explanation
Source Initiated	Sender starts the transfer	Host (Computer)	The CPU (USB host-controller) always initiates transactions.
Destination Initiated	Receiver request data	USB Host Control Bus	USB is host controlled peripherals cannot start transfers

4) Several mechanisms work together,

Handshake protocol → Confirms packet receipt or requests retransmission.

ACK → Data receive correctly

NAK → Device not ready

STALL → Device Error

CRC checks → Each packet includes CRC bits for error detection.

Interrupts / DMA → Responds to transfer completion or error events

Retry logic → Automatically resends packets if ACK not received.